WEBSITE TRAFFIC ANALYSIS

Code, Results, Visualizations & Recommendations

Dataset: traffic.csv | Date range: 19–25 August 2021

Objective

Analyse supplied music-link event logs to understand event composition, daily activity, geographic distribution, popular links/content, and practical optimization opportunities. Findings are based on the user's Colab outputs after exact-row deduplication.

Dataset overview

Item	Value
Original records	226,278
Columns	9
Exact duplicate rows detected	103,711
Rows after exact-row deduplication	122,567
Unique link IDs after cleaning	3,839
Date range	2021-08-19 to 2021-08-25

1. Data cleaning

```
df_raw = df.copy()
df_clean = df.copy()
df_clean["date"] = pd.to_datetime(df_clean["date"], errors="coerce")
df_clean = df_clean.drop_duplicates().copy()

text_columns = ["event", "country", "city", "artist", "album", "track", "isrc"]
for col in text_columns:
    df_clean[col] = df_clean[col].astype("string").str.strip().replace("", pd.NA)
df_clean["event"] = df_clean["event"].str.lower()
for col in ["country", "city", "artist", "album", "track", "isrc"]:
    df_clean[col] = df_clean[col].fillna("Unknown")
```

Output: 122,567 records remain, 103,711 exact duplicate rows were removed, and no missing values remain after filling descriptive fields.

Data-quality caution: identical rows may represent either repeated genuine events or duplicate logging. Results are deduplicated-record analysis, not definitive original traffic volume.

2. Key performance indicators

Metric	Result
Cleaned event records	122,567
Pageviews	73,360 (59.85%)
Clicks	32,499 (26.52%)

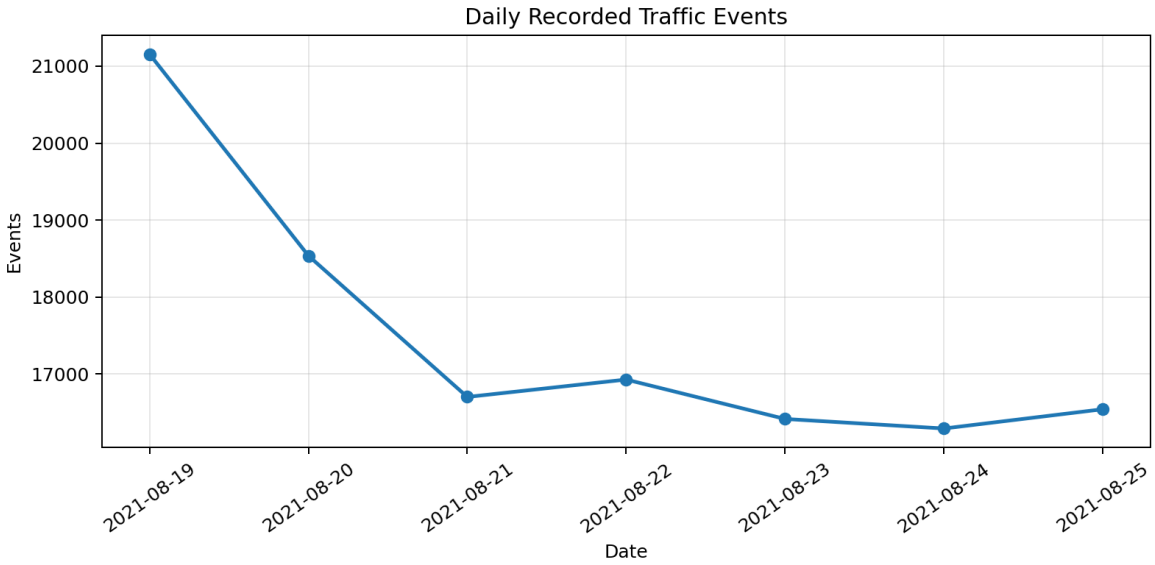
Metric	Result
Previews	16,708 (13.63%)
Average recorded events per day	17,509.57
Clicks per 100 recorded pageviews	44.30
Previews per 100 recorded pageviews	22.78

Code example

```
event_counts = df_clean["event"].value_counts()
total_events = len(df_clean)
event_summary = event_counts.rename_axis("Event").reset_index(name="Event Count")
event_summary["Percentage"] = (event_summary["Event Count"] / total_events * 100).round(2)
```

The click/pageview and preview/pageview figures are event-count ratios, not user-level conversion rates.

3. Daily traffic

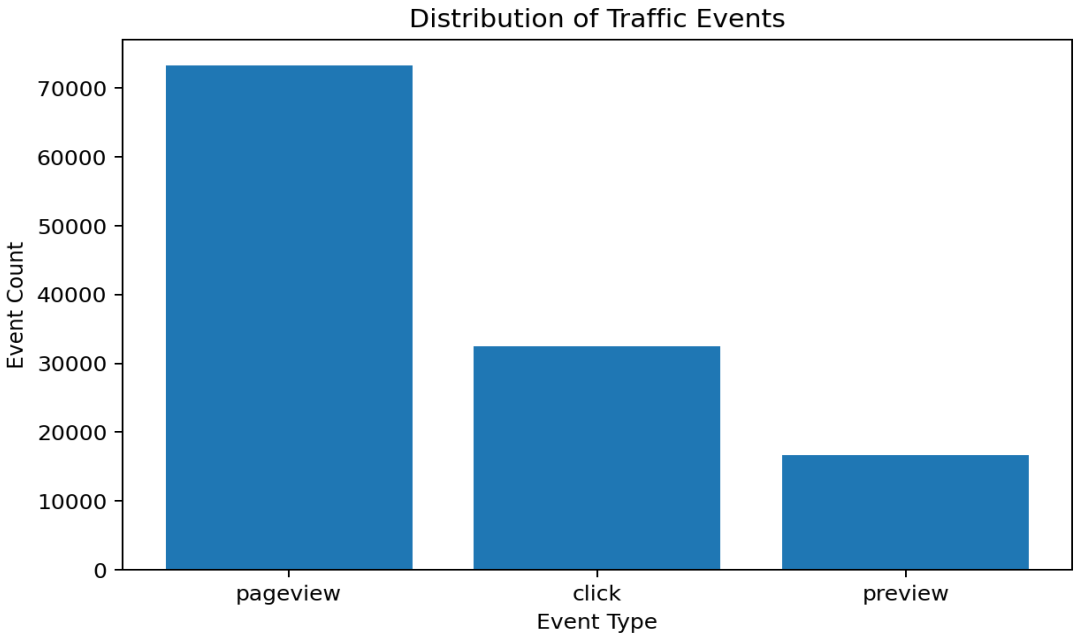


Recorded activity peaked on 19 August (21,156 events), declined over the next two days, and remained in a narrower range through 25 August. The data does not explain the cause.

Daily event totals

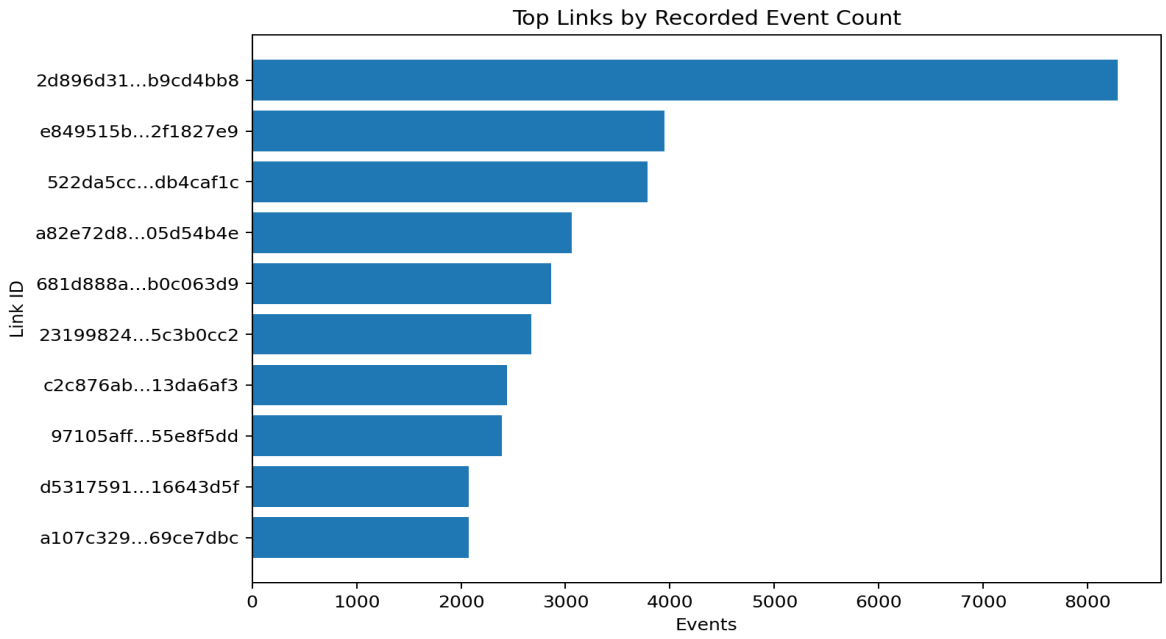
Date	Pageview	Click	Preview	Total
2021-08-19	13,006	5,860	2,290	21,156
2021-08-20	11,088	4,999	2,449	18,536
2021-08-21	9,939	4,279	2,483	16,701
2021-08-22	9,982	4,421	2,524	16,927
2021-08-23	9,722	4,314	2,379	16,415
2021-08-24	9,705	4,231	2,355	16,291
2021-08-25	9,918	4,395	2,228	16,541

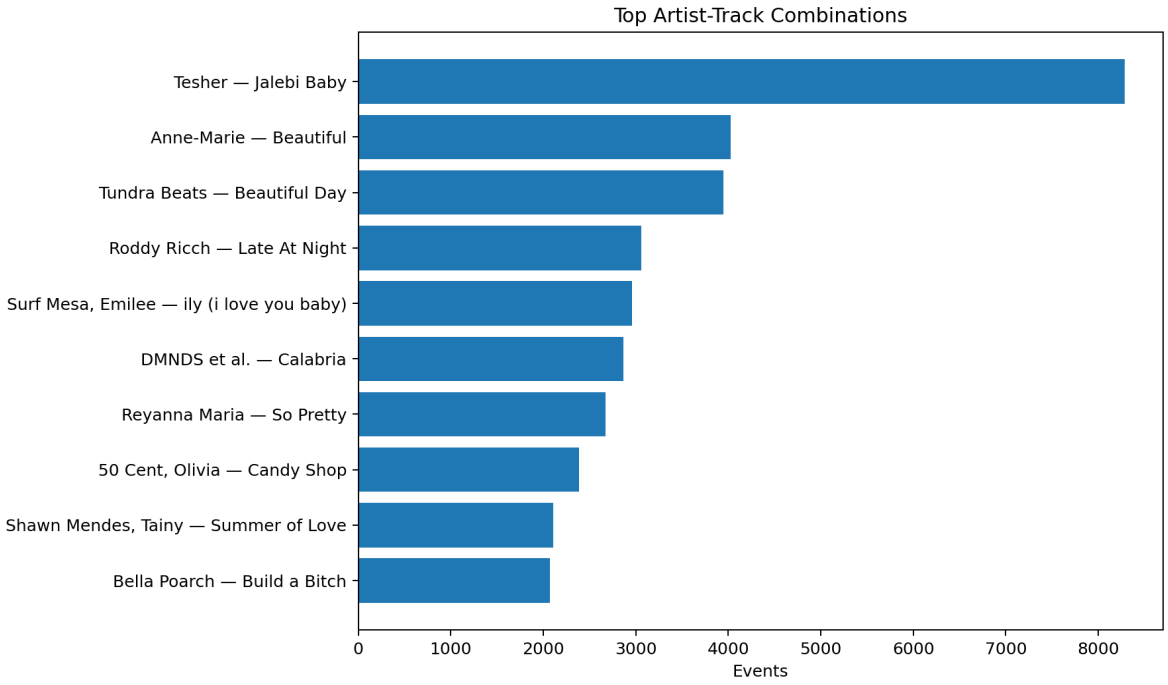
4. Event distribution



Pageviews account for 59.85% of cleaned records, clicks 26.52%, and previews 13.63%. These are shares of event rows after deduplication.

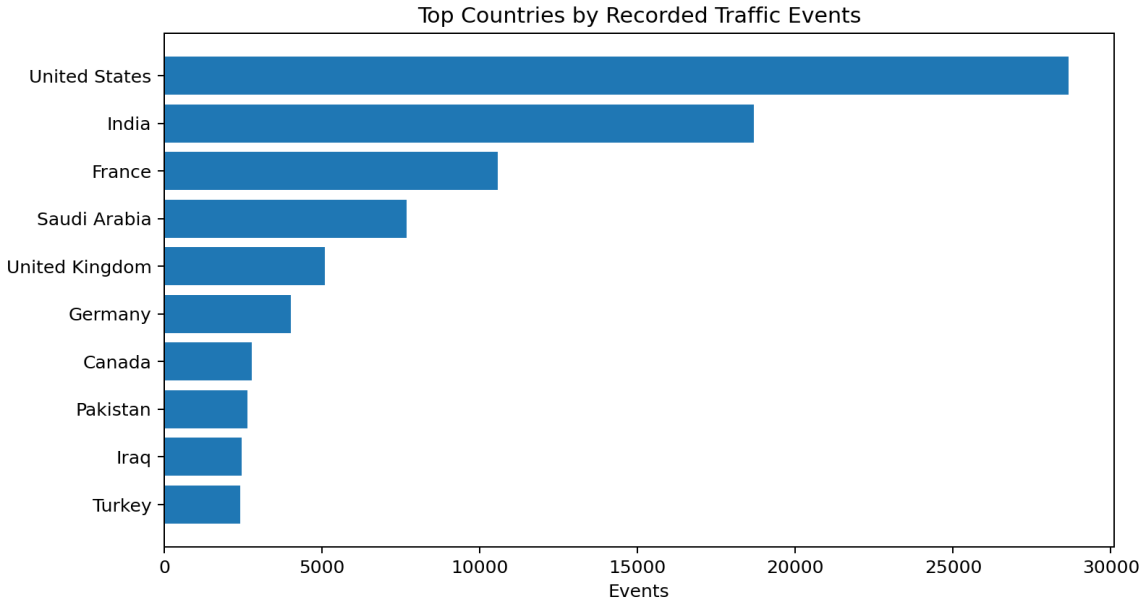
5. Top links and content





Tesher — “Jalebi Baby” is the leading artist-track combination with 8,288 recorded events. This represents event volume, not unique listeners or confirmed conversions.

6. Geographic analysis



Country	Events	Unique links
United States	28,664	2,895
India	18,689	408
France	10,565	361
Saudi Arabia	7,682	455
United Kingdom	5,095	312

Country	Events	Unique links
Germany	4,017	283
Canada	2,784	448
Pakistan	2,633	237
Iraq	2,444	230
Turkey	2,399	242

The United States recorded the most events (28,664), followed by India (18,689) and France (10,565). Counts are events, not unique people.

7. Five optimization recommendations

1. Improve call-to-action visibility — Review CTA wording, link placement, and mobile usability; compare changes using controlled tests.

2. Promote high-performing content — Test featuring popular music links, beginning with the leading artist-track combination.

3. Localize by market — Test localized messaging and music recommendations for major markets such as the United States and India.

4. Simplify the next step after previews — Make the action after a preview clear and easy to follow, then measure outcomes with improved tracking.

5. Improve analytics instrumentation — Add privacy-conscious anonymous session IDs, event timestamps, destination URLs, and campaign/source data to support real funnel analysis.

8. Limitations

• No user IDs or session IDs: unique users and sessions cannot be calculated.

• Only calendar dates are available: average session duration cannot be calculated.

• No page URLs or entry/exit fields: entry pages, exit pages, and actual user paths cannot be identified.

• Bounce rate cannot be calculated reliably from these columns.

• Exact-row deduplication removed 103,711 rows; some identical records may be legitimate repeated events.

• Event-count ratios must not be interpreted as user-level conversion rates.

Conclusion

The supplied data supports event composition, daily activity, popular content, and geographic analysis. Session-level instrumentation is needed to measure the requested session count, users, bounce rate, average duration, and entry/exit paths reliably.